

3-Adic Interpretation of Non-Trivial Collatz Cycle

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Abstract

The Collatz conjecture proposes that iterating the function $3x + 1$ for odd integers and $x/2$ for even integers always reaches 1 . This paper presents a formal refutation of non-trivial integer cycles for the accelerated Syracuse operator. We derive a Diophantine cycle equation and establish the trivial **1-4-2-1** loop as a state of equilibrium characterized by uniform division exponents $a_i = 2$. For any hypothetical cycle originating at $x > 1$, the sequence must deviate from this equilibrium. We evaluate these deviations as perturbed states and derive an ideal cycle equation. The analysis demonstrates that the **3**-adic alignment required for cycle closure and the integrality of the resulting cycle ratio cannot be simultaneously maintained across the partitioned operational domains. Consequently, no non-trivial positive integer cycle can occur.

Keywords: Collatz conjecture, $3x + 1$ problem, Diophantine equations, **3**-adic valuation

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1 Introduction

The Collatz conjecture, stated by Lothar Collatz in 1937, concerns the behavior of the function $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ defined by

$$f(n) = \begin{cases} 3n + 1, & \text{if } n \equiv 1 \pmod{2}, \\ \frac{n}{2}, & \text{if } n \equiv 0 \pmod{2}. \end{cases} \quad (1)$$

The conjecture asserts that for any starting integer n , the sequence of iterates $f^k(n)$ will eventually reach the integer 1. Despite its elementary statement, a formal proof has remained one of the most elusive problems in mathematics [5].

Notable prior works are summarized as follows.

1. The study of non-trivial integer cycles in the Collatz conjecture is centered on a fundamental Diophantine balance between the growth factor 3^n and the decay factor 2^n . In 1977, Ray Steiner [8] provided the first major breakthrough by rigorously excluding the existence of non-trivial “1-cycles” (circuits with a single local minimum) in the positive integers. By applying Baker’s theorem on linear forms in logarithms, Steiner proved that the ratio of rising to falling steps must satisfy extremely rare rational approximations of $\log_2 3$, which are impossible for non-trivial integers.
2. Research on generalized “ m -cycles”—periodic orbits characterized by m local minima—was established by John Simons and Benne de Weger [7]. Using transcendence theory and Rhin’s estimates for linear forms in logarithms, they proved that any hypothetical non-trivial cycle must possess a large number of “peaks” and “valleys”. Their work initially confirmed the non-existence of non-trivial cycles for $m \leq 68$ (later updated to $m \leq 75$), and recent refinements by Christian Hercher [3] have extended this barrier to $m \geq 92$.
3. Shalom Eliahou [2] introduced a criterion based on parity vectors and the “ones-ratio” theorem, showing that the fraction of odd terms in a cycle must be a convergent of the continued fraction expansion of $\log_2 3$. This mathematical framework allows researchers to translate computational verification limits into rigorous lower bounds for cycle length. Eliahou’s method established that even at modest verification thresholds, any non-trivial cycle would be forced to contain millions or billions of elements.
4. High-performance computational projects, most notably those by David Barina [1], have verified the conjecture for all integers up to 2^{71} , approximately 2.36×10^{21} . This exhaustive verification implies that a non-trivial cycle, if it exists, must have a total period length exceeding 355,504,839,929 steps. Furthermore, recent 2025 research by Jochen Kiemes [4] leveraging bit-position dynamics has derived a substantial structural lower bound of at least 17,026,679,261 steps for any hypothetical cycle.
5. While non-trivial cycles are excluded in the positive domain up to these massive limits, the negative integer domain contains three confirmed non-trivial cycles at -1 , -5 , and -17 . These known loops serve as critical benchmarks for contemporary structural proofs, such as Lionel Laureore’s 2025 modulo 12 analysis [6]. These modern approaches attempt to demonstrate that deterministic residue class transitions preclude any additional periodicities, suggesting that the trivial $\{4, 2, 1\}$ cycle is indeed the only attractor for all positive integers.

2 Theoretical overview

The proof is organized through the following sequence of reductions and exclusions.

1. **Iterative expansion.** Section 3 defines the accelerated odd-to-odd operator

$$T(x) = \frac{3x+1}{2^a}$$

and iterates it to construct the multi-step numerator sequence:

$$2^v k_{n+1} + m_{n+1} = \frac{3^n 2^v k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + a_2 + \dots + a_n}}.$$

2. **The fundamental cycle equation.** Equating the terminal iterated value to the initial value gives

$$Z = \frac{3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{S_j}}{2^S - 3^n} = \frac{N}{D}.$$

3. **Lemma 1A and the equilibrium state.** The baseline trajectory with every division exponent $a_i = 2$ is evaluated. This state forces the Diophantine solution $Z = 1$, isolating the trivial loop.
4. **The perturbation model.** The deviation metric $\delta_i = a_i - 2$ and the cumulative deviation S'_j are introduced to quantify trajectories departing from the equilibrium baseline.
5. **Lemma 1B (Exact difference formula).** The exact rational difference between a perturbed numerator sum and the equilibrium numerator sum is expressed through the perturbation factors $2^{S'_j} - 1$.
6. **Lemma 1C (Pure negative configuration).** Trajectories containing only non-positive perturbations, $\delta_i \leq 0$, are evaluated. The resulting magnitude deficit forces the actual numerator to remain too large relative to the denominator.
7. **Lemma 1D (Pure positive configuration).** Trajectories containing only non-negative perturbations, $\delta_i \geq 0$, are evaluated. Exponential accumulation forces the denominator to outgrow the numerator, so that $N_{\text{new}} < D_{\text{new}}$.
8. **Lemma 2D (Modified cycle equation).** The required cycle equation is rewritten as
$$\Delta N_{\text{required}} = (Z 2^S - 2^{2^n}) + (1 - Z) 3^n.$$
9. **lemma 3D** Non-equilibrium equality of ΔN_{actual} and ΔD is impossible.
10. **Lemma 1E (Parity lemma).** The parity of the cumulative exponent deviation d_{k+1} is inverted exactly when the preceding operational step a_k is odd.
11. **Lemma 1F (LHS bifurcation and 3-adic constraints).** The modified cycle equation is bifurcated into an internal sequence summation and a terminal boundary term T_{last} . Cycle closure requires the terminal valuation condition

$$v_3(T_{\text{last}}) \geq 1.$$

12. **Lemma 1G (LTE lift calculation)**. The Lifting The Exponent lemma is invoked to calculate the minimum deviation

$$d_k = 2 \cdot 3^{p-n+k}$$

required to satisfy the internal 3-adic divisibility condition.

13. **Lemma 1H (Permissible trajectories)**. The possible 3-adic alignment mechanisms are classified into Mechanism 1, direct lifting, and Mechanism 2, cancellation-lift. The endpoint $p = 1$ gives the no-lift pure negative branch, while the lifted range is routed to Lemmas 1J and 1K.
14. **Lemma 1I**. The pure negative no-lift branch is exhausted through its terminal subcases, and each case forces a non-integral cycle ratio Z .
15. **Lemma 1J**. The minimal lifted-tail mechanism is shown to force the ratio bound $Z < 5$.
16. **Lemma 1K**. Positive exponent inflation further decreases the cycle ratio.
17. **Theorem 1 (Synthesis)**. The perturbation, trajectory, and terminal valuation exhaustions are combined to rule out every non-trivial positive integer cycle.

3 Theory

3.1 Multiple iterations on modular classes

We define the specific Collatz iteration terms as follows:

- (a) I^1 represents a single $3x + 1$ operation on a starting odd integer N_0 .
- (b) I^2 represents division by 2^a , where $a = v_2(3N_0 + 1)$.
- (c) I^d represents one application of I^1 followed by one application of I^2 .

Let a sequence have n allowed transformations, with division exponents

$$a_1, a_2, \dots, a_n.$$

The operation I^d applied to $2^v k_1 + m_1$ yields the following successive odd integers:

$$\begin{aligned} 2^v k_1 + m_1 &\longrightarrow \frac{3 \cdot 2^v k_1 + 3m_1 + 1}{2^{a_1}}, \\ &\longrightarrow \frac{3^2 \cdot 2^v k_1 + 3^2 m_1 + 3 + 2^{a_1}}{2^{a_1+a_2}}, \\ &\longrightarrow \frac{3^3 \cdot 2^v k_1 + 3^3 m_1 + 3^2 + 3 \cdot 2^{a_1} + 2^{a_1+a_2}}{2^{a_1+a_2+a_3}}, \\ &\longrightarrow \frac{3^4 \cdot 2^v k_1 + 3^4 m_1 + 3^3 + 3^2 \cdot 2^{a_1} + 3 \cdot 2^{a_1+a_2} + 2^{a_1+a_2+a_3}}{2^{a_1+a_2+a_3+a_4}}. \end{aligned}$$

In general, after n steps, if the integer becomes $2^v k_{n+1} + m_{n+1}$, then

$$2^v k_{n+1} + m_{n+1} =$$

$$\frac{3^n 2^v k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + a_2 + \dots + a_n}}. \quad (2)$$

If the trajectory closes after n applications of I^d , then

$$2^v k_{n+1} + m_{n+1} = 2^v k_1 + m_1.$$

Thus (2) becomes

$$2^v k_1 + m_1 = \frac{3^n 2^v k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + a_2 + \dots + a_n}}. \quad (3)$$

This is the fundamental linear Diophantine equation representing an integer loop.

Lemma 1A *If all division exponents satisfy $a_i = p$, then the only possible integer cycle is the trivial cycle $4 \rightarrow 2 \rightarrow 1 \rightarrow 4$, occurring when $p = 2$.*

Proof Let $n \geq 1$ be the length of a hypothetical cycle, and let

$$a = (a_1, \dots, a_n), \quad a_i \in \mathbb{Z}^+.$$

For $0 \leq j \leq n$, define

$$S_j = \sum_{i=1}^j a_i, \quad S_0 = 0,$$

and set

$$S = S_n = \sum_{i=1}^n a_i.$$

The numerator and denominator of the cycle ratio are

$$N = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}, \quad D = 2^S - 3^n,$$

and

$$Z = \frac{N}{D}.$$

Starting from (3), multiplying both sides by 2^S gives

$$2^v k_1 2^S + m_1 2^S = 3^n 2^v k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{S_{n-1}}.$$

Grouping the k_1 and m_1 terms yields

$$2^v k_1 (2^S - 3^n) = m_1 (3^n - 2^S) + 3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{S_{n-1}}.$$

Hence

$$\begin{aligned} k_1 &= -\frac{m_1}{2^v} + \frac{3^{n-1} + 3^{n-2} 2^{a_1} + \dots + 2^{S_{n-1}}}{2^v (2^S - 3^n)} \\ &= \frac{1}{2^v} (Z - m_1), \end{aligned}$$

where

$$Z = \frac{3^{n-1} + 3^{n-2}2^{a_1} + \dots + 2^{S_{n-1}}}{2^S - 3^n} = \frac{N}{D}.$$

Now assume that all division exponents are equal to a fixed positive integer p :

$$a_i = p \quad (1 \leq i \leq n).$$

Then

$$S_j = jp$$

for every j , and therefore

$$N = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{jp}.$$

This is a finite geometric progression:

$$N = 3^{n-1} + 3^{n-2}2^p + 3^{n-3}(2^p)^2 + \dots + (2^p)^{n-1},$$

with first term 3^{n-1} and common ratio $2^p/3$. Therefore

$$N = 3^{n-1} \left[\frac{\left(\frac{2^p}{3}\right)^n - 1}{\frac{2^p}{3} - 1} \right] = \frac{2^{np} - 3^n}{2^p - 3}.$$

Since $S = np$, the denominator is

$$D = 2^{np} - 3^n.$$

Thus

$$Z = \frac{N}{D} = \frac{1}{2^p - 3}.$$

For Z to be a positive integer, one must have

$$2^p - 3 = 1.$$

Hence $2^p = 4$, so $p = 2$, and consequently $Z = 1$.

Substituting $Z = 1$ into

$$k_1 = \frac{1}{2^v}(Z - m_1)$$

gives

$$k_1 = \frac{1 - m_1}{2^v}.$$

Since k_1 is a non-negative integer and m_1 is a positive odd residue, the only possible value is

$$m_1 = 1, \quad k_1 = 0.$$

Thus the corresponding odd integer is 1, which gives the ordinary Collatz loop

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

Therefore the only possible uniform-exponent integer cycle is the trivial cycle. $\square$

Corollary 1A-1 *For Z to be a positive integer, the denominator*

$$D = 2^S - 3^n$$

must be positive. Hence

$$2^S > 3^n,$$

or equivalently

$$\frac{S}{n} > \log_2 3.$$

Thus the threshold condition is

$$\frac{S}{n} > 1.585, \quad S > 1.585 n.$$

3.2 The perturbation model

By Lemma 1A, among uniform exponent vectors no integer solution is possible for Z unless $a_i = 2$. We therefore take the equilibrium state

$$N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$$

as the baseline. Every possible exponent vector is then examined by increasing or decreasing the division exponents from this equilibrium position.

The perturbation classes are:

- **Case I:** pure increment, where the exponents from a_1 to a_n receive only positive perturbations;
- **Case II:** pure decrement, where the exponents from a_1 to a_n receive only negative perturbations;
- **Case III:** mixed perturbations with net increment in the total exponent sum;
- **Case IV:** mixed perturbations with net decrement in the total exponent sum;
- **Case V:** mixed perturbations with no net alteration in the total exponent sum.

These cases are tracked through the magnitude of the perturbed numerator relative to the equilibrium numerator. A perturbed trajectory may increase the numerator, decrease the numerator, or, by cancellation among mixed perturbations, preserve the equilibrium numerator.

Lemma 1B *For any valid perturbation of the division exponents from equilibrium, exactly one of the following alternatives holds:*

$$N_{\text{new}} > N_{\text{eq}}, \quad N_{\text{new}} < N_{\text{eq}}, \quad N_{\text{new}} = N_{\text{eq}}.$$

The equality case may occur trivially at equilibrium or non-trivially through mixed-prefix cancellation.

Proof Let $n \geq 1$ be fixed, and let

$$a = (a_1, \dots, a_n), \quad a_i \geq 1.$$

Define the perturbation from equilibrium by

$$\delta_i = a_i - 2,$$

and define the prefix perturbation sum by

$$S'_j = \sum_{i=1}^j \delta_i, \quad S'_0 = 0.$$

Let

$$s_j = \sum_{i=1}^j a_i.$$

Then

$$s_j = 2j + S'_j.$$

The perturbed numerator is

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{s_j},$$

while the equilibrium numerator is

$$N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{2j}.$$

Substituting $s_j = 2j + S'_j$ gives

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{2j} 2^{S'_j}.$$

Hence

$$\begin{aligned} N_{\text{new}} - N_{\text{eq}} &= \sum_{j=0}^{n-1} 3^{n-1-j} 2^{2j} (2^{S'_j} - 1) \\ &= \sum_{j=1}^{n-1} 3^{n-1-j} 2^{2j} (2^{S'_j} - 1), \end{aligned}$$

since the $j = 0$ term vanishes.

Let

$$w_j = 3^{n-1-j} 2^{2j}.$$

Each w_j is positive. Therefore

$$N_{\text{new}} - N_{\text{eq}} = \sum_{j=1}^{n-1} w_j (2^{S'_j} - 1).$$

If $S'_j \geq 0$ for all j and at least one $S'_j > 0$, then every summand is non-negative and at least one is positive. Thus

$$N_{\text{new}} > N_{\text{eq}}.$$

If $S'_j \leq 0$ for all j and at least one $S'_j < 0$, then every summand is non-positive and at least one is negative. Thus

$$N_{\text{new}} < N_{\text{eq}}.$$

In the mixed-prefix case, positive and negative summands may compete, so the resulting value is one of

$$N_{\text{new}} > N_{\text{eq}}, \quad N_{\text{new}} < N_{\text{eq}}, \quad N_{\text{new}} = N_{\text{eq}}.$$

This exhausts the possible magnitude outcomes. □

3.3 Constraints in the perturbation model

The perturbation model is governed by the following constraints.

Constraint 1: Division exponent lower bound. The division exponent is

$$a_i = v_2(3x_{i-1} + 1).$$

Since x_{i-1} is a positive odd integer, $3x_{i-1} + 1$ is a positive even integer. Therefore

$$a_i \geq 1$$

for every i . Consequently, the only possible negative perturbation from the equilibrium value 2 is

$$\delta_i = -1.$$

Thus no division exponent can be lowered more than once.

Constraint 2: Modulo 3 incompatibility. Both D and N , in perturbed and unperturbed form, are odd and not divisible by 3.

Constraint 3: Ratio bounds. If an integer cycle forms, then

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}} \geq 5$$

for all cases with $N_{\text{new}} > D_{\text{new}}$, while

$$\frac{N_{\text{new}}}{D_{\text{new}}} = 1$$

in the possible equality case $N_{\text{new}} = D_{\text{new}}$.

Lemma 1C Assume $n \geq 1$ and let the exponent vector $a = (a_1, \dots, a_n)$ satisfy

$$1 \leq a_i \leq 2$$

for every i . Assume further that the perturbation is non-trivial, so at least one exponent satisfies $a_i = 1$. Let

$$S = \sum_{i=1}^n a_i,$$

and define

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}, \quad D_{\text{new}} = 2^S - 3^n.$$

Then

$$N_{\text{new}} > D_{\text{new}}.$$

Proof Let p be the number of negative perturbations, equivalently the number of indices for which $a_i = 1$. Since the perturbation is non-trivial,

$$p \geq 1.$$

The total exponent sum is

$$S = 2n - p.$$

For $0 \leq j \leq n-1$, the remaining exponent sum after index j satisfies

$$S - S_j = \sum_{i=j+1}^n a_i \leq 2(n-j),$$

because each $a_i \leq 2$. Hence

$$S_j \geq S - 2(n-j).$$

Substituting this lower bound into the numerator gives

$$\begin{aligned} N_{\text{new}} &= \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j} \\ &\geq \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S-2(n-j)}. \end{aligned}$$

Factoring out 2^S gives

$$N_{\text{new}} \geq 2^S \sum_{j=0}^{n-1} 3^{n-1-j} 4^{-(n-j)}.$$

Writing $m = n - j$, this becomes

$$N_{\text{new}} \geq 2^S \sum_{m=1}^n 3^{m-1} 4^{-m}.$$

The sum is a finite geometric progression, and therefore

$$\sum_{m=1}^n 3^{m-1} 4^{-m} = 1 - \left(\frac{3}{4}\right)^n.$$

Thus

$$N_{\text{new}} \geq 2^S \left[1 - \left(\frac{3}{4}\right)^n \right].$$

Now compare this lower bound with the denominator:

$$\begin{aligned} N_{\text{new}} - D_{\text{new}} &\geq 2^S \left[1 - \left(\frac{3}{4}\right)^n \right] - (2^S - 3^n) \\ &= 3^n - 2^S \left(\frac{3}{4}\right)^n. \end{aligned}$$

Using $S = 2n - p$, we obtain

$$\begin{aligned} N_{\text{new}} - D_{\text{new}} &\geq 3^n - 2^{2n-p} \left(\frac{3}{4}\right)^n \\ &= 3^n - 3^n 2^{-p} \\ &= 3^n \left(1 - \frac{1}{2^p} \right). \end{aligned}$$

Since $p \geq 1$, the factor $1 - 2^{-p}$ is positive. Therefore

$$N_{\text{new}} - D_{\text{new}} > 0,$$

and hence

$$N_{\text{new}} > D_{\text{new}}.$$

□

Lemma 1D *Any purely positive perturbation of the division exponents from equilibrium forces*

$$D_{\text{new}} > N_{\text{new}}.$$

Consequently, no purely positive perturbation can satisfy the positive integer cycle condition.

Proof Let

$$a = (a_1, \dots, a_n)$$

be the perturbed exponent vector, and let

$$\delta_i = a_i - 2.$$

For a purely positive perturbation, every $\delta_i \geq 0$, and at least one $\delta_i > 0$. Define

$$S'_j = \sum_{i=1}^j \delta_i, \quad S'_0 = 0.$$

Then

$$S'_n \geq 1.$$

Let N_{eq} and D_{eq} denote the equilibrium numerator and denominator. By Lemma 1A,

$$N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n.$$

Let

$$\Delta N = N_{\text{new}} - N_{\text{eq}}, \quad \Delta D = D_{\text{new}} - D_{\text{eq}}.$$

It is enough to prove

$$\Delta D > \Delta N.$$

The total exponent sum is

$$S = 2n + S'_n.$$

Therefore

$$\begin{aligned} \Delta D &= (2^{2n+S'_n} - 3^n) - (2^{2n} - 3^n) \\ &= 2^{2n} (2^{S'_n} - 1). \end{aligned}$$

By Lemma 1B,

$$\Delta N = \sum_{j=1}^{n-1} 3^{n-1-j} 2^{2j} (2^{S'_j} - 1).$$

Since every $\delta_i \geq 0$, the prefix sums satisfy

$$0 = S'_0 \leq S'_1 \leq \dots \leq S'_{n-1} \leq S'_n.$$

Hence

$$2^{S'_j} - 1 \leq 2^{S'_n} - 1$$

for every j . Thus

$$\begin{aligned} \Delta N &\leq (2^{S'_n} - 1) \sum_{j=1}^{n-1} 3^{n-1-j} 2^{2j} \\ &< (2^{S'_n} - 1) N_{\text{eq}}. \end{aligned}$$

Since

$$N_{\text{eq}} = 2^{2n} - 3^n < 2^{2n}$$

and $2^{S'_n} - 1 > 0$, it follows that

$$\Delta N < (2^{S'_n} - 1) 2^{2n} = \Delta D.$$

Therefore

$$D_{\text{new}} > N_{\text{new}}.$$

This violates the positive integer cycle condition. $\square$

Constraint 4: Formulation of actual ΔN . The perturbed numerator is

$$N_{\text{new}} = 3^{n-1} + 3^{n-2}2^{S_1} + 3^{n-3}2^{S_2} + \dots + 2^{S_{n-1}},$$

where $S_1, S_2, \dots, S_{n-1}$ are the perturbed prefix sums of exponents. The equilibrium numerator is

$$N_{\text{eq}} = 3^{n-1} + 3^{n-2}2^2 + 3^{n-3}2^4 + \dots + 2^{2n-2}.$$

If

$$N_{\text{eq}} > N_{\text{new}},$$

then

$$\Delta N_{\text{actual}} = N_{\text{eq}} - N_{\text{new}},$$

and the k -th term is

$$\text{Term}_k = 3^{n-k} (2^{2k-2} - 2^{S_{k-1}}).$$

Conversely, if

$$N_{\text{new}} > N_{\text{eq}},$$

then

$$\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}},$$

and the k -th term is

$$\text{Term}_k = 3^{n-k} (2^{S_{k-1}} - 2^{2k-2}).$$

Constraint 5: Bounds on negative perturbations. The number of pure negative perturbations p must satisfy

$$p \geq 1$$

and

$$p < 0.415 n.$$

This follows from the threshold condition

$$S > 1.585 n.$$

Since

$$S = 2n - p$$

in the pure negative case, we have

$$2n - p > 1.585 n,$$

and therefore

$$p < (2 - 1.585)n = 0.415 n.$$

Similarly, any total decrement in a mixed case must preserve this bound in order to keep

$$D_{\text{new}} > 0.$$

Constraint 6: Formulation of required ΔN . This condition defines the exact value of the net numerator change

$$\Delta N_{\text{required}}$$

required for a cycle with a fixed integer ratio $Z \geq 5$ to exist. Its explicit form is derived in Lemma 2D.

Lemma 2D *Let*

$$N_{\text{eq}} = 2^{2n} - 3^n, \quad D_{\text{new}} = 2^S - 3^n,$$

and suppose that a candidate cycle has integer ratio

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}}.$$

Then cycle closure forces the actual numerator displacement ΔN_{actual} to equal the required numerator displacement $\Delta N_{\text{required}}$.

In the net-decrement case

$$N_{\text{new}} = N_{\text{eq}} - \Delta N_{\text{actual}},$$

one has

$$\Delta N_{\text{actual}} = \Delta N_{\text{required}} = (2^{2n} - Z2^S) + (Z - 1)3^n.$$

In the net-increment case

$$N_{\text{new}} = N_{\text{eq}} + \Delta N_{\text{actual}},$$

one has

$$\Delta N_{\text{actual}} = \Delta N_{\text{required}} = (Z2^S - 2^{2n}) + (1 - Z)3^n.$$

Proof The cycle condition is

$$N_{\text{new}} = ZD_{\text{new}}.$$

First assume net decrement, so that

$$N_{\text{new}} = N_{\text{eq}} - \Delta N_{\text{actual}}.$$

Then

$$N_{\text{eq}} - \Delta N_{\text{actual}} = ZD_{\text{new}},$$

and hence

$$\Delta N_{\text{actual}} = N_{\text{eq}} - ZD_{\text{new}}.$$

Substituting

$$N_{\text{eq}} = 2^{2n} - 3^n, \quad D_{\text{new}} = 2^S - 3^n,$$

gives

$$\Delta N_{\text{actual}} = (2^{2n} - 3^n) - Z(2^S - 3^n) = (2^{2n} - Z2^S) + (Z - 1)3^n.$$

The net-increment case is analogous. If

$$N_{\text{new}} = N_{\text{eq}} + \Delta N_{\text{actual}},$$

then

$$N_{\text{eq}} + \Delta N_{\text{actual}} = ZD_{\text{new}},$$

so

$$\Delta N_{\text{actual}} = ZD_{\text{new}} - N_{\text{eq}}.$$

Substitution gives

$$\Delta N_{\text{actual}} = Z(2^S - 3^n) - (2^{2n} - 3^n) = (Z2^S - 2^{2n}) + (1 - Z)3^n.$$

Thus, in each branch, the perturbatively computed actual displacement must equal the algebraically required displacement forced by cycle closure. $\square$

Corollary 2D-1 *In the net-increment branch, cycle closure is equivalent to*

$$\Delta N_{\text{actual}} + B = (1 - Z)3^n,$$

where

$$B := 2^{2n} - Z2^S.$$

Consequently, the bifurcated left-hand side must satisfy the forced 3-adic valuation condition

$$v_3(\Delta N_{\text{actual}} + B) = v_3((1 - Z)3^n) = n + v_3(Z - 1).$$

Proof By Lemma 2D, in the net-increment branch,

$$\Delta N_{\text{actual}} = \Delta N_{\text{required}} = (Z2^S - 2^{2n}) + (1 - Z)3^n.$$

Moving the non-3ⁿ term to the left gives

$$\Delta N_{\text{actual}} - (Z2^S - 2^{2n}) = (1 - Z)3^n.$$

Equivalently, with

$$B := 2^{2n} - Z2^S,$$

we obtain

$$\Delta N_{\text{actual}} + B = (1 - Z)3^n.$$

Applying v_3 to both sides gives

$$v_3(\Delta N_{\text{actual}} + B) = v_3((1 - Z)3^n) = n + v_3(Z - 1).$$

The net-decrement branch is identical after replacing

$$B := 2^{2n} - Z2^S$$

by

$$B := Z2^S - 2^{2n}$$

and replacing $(1 - Z)3^n$ by $(Z - 1)3^n$; in either case the resulting right-hand side has 3-adic valuation $n + v_3(Z - 1)$. This is the required 3-adic alignment condition. $\square$

4 Mathematical consequences of perturbations

Lemma 3D 2-Adic Misalignment of the Neutral Branch. *For any non-equilibrium permutation of the equilibrium exponent profile,*

$$\Delta D \neq \Delta N_{\text{actual}}.$$

Equivalently, if

$$\Delta D = \Delta N_{\text{actual}},$$

then the exponent profile is the equilibrium profile

$$a_1 = a_2 = \dots = a_n = 2.$$

Proof Recall that

$$D_{\text{new}} = 2^{S_n} - 3^n, \quad D_{\text{eq}} = 2^{2n} - 3^n.$$

Hence

$$\Delta D = D_{\text{new}} - D_{\text{eq}} = 2^{S_n} - 2^{2n}.$$

Also,

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}, \quad N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{2j}.$$

Therefore

$$\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} (2^{S_j} - 2^{2j}).$$

The $j = 0$ term is zero, so

$$\Delta N_{\text{actual}} = \sum_{j=1}^{n-1} 3^{n-1-j} (2^{S_j} - 2^{2j}).$$

Assume first that the profile is the equilibrium profile:

$$a_1 = a_2 = \dots = a_n = 2.$$

Then

$$S_j = 2j \quad (0 \leq j \leq n),$$

and hence

$$\Delta D = 0, \quad \Delta N_{\text{actual}} = 0.$$

Conversely, assume the profile is non-equilibrium. There are two cases.

Case 1: Some numerator-visible prefix deviates. Let $t < n$ be the first index such that

$$S_t \neq 2t.$$

Then for every $j < t$,

$$S_j = 2j,$$

so all earlier terms in ΔN_{actual} vanish. The first nonzero term is

$$3^{n-1-t} (2^{S_t} - 2^{2t}).$$

Since 3^{n-1-t} is odd, its 2-adic valuation is

$$v_2(3^{n-1-t} (2^{S_t} - 2^{2t})) = \min(S_t, 2t).$$

For every later index $j > t$, we have

$$S_j > S_t, \quad 2j > 2t,$$

because every division exponent satisfies $a_i \geq 1$. Therefore

$$\min(S_j, 2j) > \min(S_t, 2t).$$

Hence no later term can cancel the lowest 2-adic layer of the first nonzero term. Consequently,

$$v_2(\Delta N_{\text{actual}}) = \min(S_t, 2t).$$

On the other hand,

$$\Delta D = 2^{S_n} - 2^{2n}.$$

Since the profile is non-equilibrium, if $S_n \neq 2n$, then

$$v_2(\Delta D) = \min(S_n, 2n).$$

Because $t < n$, we have

$$S_t < S_n, \quad 2t < 2n.$$

Thus

$$\min(S_t, 2t) < \min(S_n, 2n).$$

Therefore

$$v_2(\Delta N_{\text{actual}}) < v_2(\Delta D).$$

So

$$\Delta N_{\text{actual}} \neq \Delta D.$$

If instead $S_n = 2n$, then

$$\Delta D = 0.$$

But the t -th term above is nonzero and its lowest 2-adic layer cannot be canceled by any later term. Hence

$$\Delta N_{\text{actual}} \neq 0 = \Delta D.$$

Again,

$$\Delta N_{\text{actual}} \neq \Delta D.$$

Case 2: No numerator-visible prefix deviates. Then

$$S_j = 2j \quad (0 \leq j \leq n-1).$$

Since the profile is still assumed non-equilibrium, the only possible deviation is in the final exponent a_n . Hence

$$S_n \neq 2n.$$

Therefore

$$\Delta N_{\text{actual}} = 0,$$

while

$$\Delta D = 2^{S_n} - 2^{2n} \neq 0.$$

So

$$\Delta N_{\text{actual}} \neq \Delta D.$$

Thus every non-equilibrium profile satisfies

$$\Delta D \neq \Delta N_{\text{actual}}.$$

Equivalently, the identity

$$\Delta D = \Delta N_{\text{actual}}$$

is possible only in the equilibrium profile

$$a_1 = a_2 = \dots = a_n = 2.$$

□

Lemma 1E *The parity of the exponent deviation d_{k+1} is inverted if and only if the preceding perturbation-state step a_k is odd. This includes a negative perturbation $a_k = 1$ and any odd positive perturbation.*

Proof Let S_{k-1} denote the actual prefix sum of the exponents before step k , and let

$$2k - 2$$

be the corresponding equilibrium prefix sum. Define

$$d_k = |S_{k-1} - (2k - 2)|.$$

Since

$$S_k = S_{k-1} + a_k,$$

we have

$$\begin{aligned} d_{k+1} &= |S_k - 2k| \\ &= |(S_{k-1} + a_k) - ((2k - 2) + 2)| \\ &= |\{S_{k-1} - (2k - 2)\} + (a_k - 2)|. \end{aligned}$$

The absolute value does not change parity, since $|x| \equiv x \pmod{2}$. Therefore

$$d_{k+1} \equiv (S_{k-1} - (2k - 2)) + (a_k - 2) \pmod{2}.$$

Since

$$d_k \equiv S_{k-1} - (2k - 2) \pmod{2},$$

we obtain

$$d_{k+1} \equiv d_k + a_k - 2 \pmod{2}.$$

Equivalently,

$$d_{k+1} \equiv d_k + a_k \pmod{2}.$$

If a_k is even, then $a_k \equiv 0 \pmod{2}$, so

$$d_{k+1} \equiv d_k \pmod{2}.$$

Thus even steps preserve parity.

If a_k is odd, then $a_k \equiv 1 \pmod{2}$, so

$$d_{k+1} \equiv d_k + 1 \pmod{2}.$$

Thus odd steps invert parity. □

Lemma 1F *The modified cycle equation bifurcates into internal summation terms and a terminal boundary block. It forces a non-zero terminal boundary valuation,*

$$v_3(T_{\text{last}}) \geq 1,$$

and forbids any below-target term from acting as a unique minimum 3-adic valuation.

Proof In the net-increment form, write the actual numerator change as

$$\Delta N_{\text{actual}} = \sum_{k=2}^{n-1} T_k + T_n,$$

where

$$T_k = 3^{n-k} \left(2^{S_{k-1}} - 2^{2k-2} \right),$$

and in particular

$$T_n = 3^{n-n} \left(2^{S_{n-1}} - 2^{2n-2} \right) = 2^{S_{n-1}} - 2^{2n-2}.$$

By Lemma 2D, in the net-increment convention,

$$\Delta N_{\text{actual}} = (Z2^S - 2^{2n}) + (1 - Z)3^n.$$

Thus, with the net-increment boundary term

$$B := 2^{2n} - Z2^S,$$

we have

$$\Delta N_{\text{actual}} + B = (1 - Z)3^n.$$

Define

$$T_{\text{last}} := T_n + B.$$

Then the cycle equation becomes

$$\left(\sum_{k=2}^{n-1} T_k \right) + T_{\text{last}} = (1 - Z)3^n.$$

Hence the required target valuation is

$$v_3((1 - Z)3^n) = n + v_3(1 - Z).$$

Since

$$1 - Z = -(Z - 1)$$

and $v_3(-x) = v_3(x)$, this is equivalently

$$v_3((Z - 1)3^n) = n + v_3(Z - 1).$$

For every internal term T_k with $2 \leq k \leq n - 1$, the factor 3^{n-k} gives

$$v_3(T_k) \geq 1$$

unless the parenthetical factor is zero. Therefore the internal summation is divisible by 3. Since the right-hand side is divisible by 3, the terminal block must also be divisible by 3. If

$$v_3(T_{\text{last}}) = 0,$$

then the full left-hand side has valuation 0, contradicting the right-hand side valuation. Thus

$$v_3(T_{\text{last}}) \geq 1.$$

Now suppose that some internal term T_j with $j < n$ has valuation exactly equal to its visible positional valuation:

$$v_3(T_j) = n - j,$$

and suppose this valuation is strictly smaller than the valuation of every other active component. Then T_j is a unique minimum. By the non-archimedean valuation rule,

$$v_3\left(\sum T_k\right) = \min_k v_3(T_k) = v_3(T_j) = n - j.$$

But

$$n - j < n \leq n + v_3(Z - 1),$$

so the full left-hand side cannot reach the cycle-forced target valuation. This is impossible.

Therefore no active component below the target may remain as a unique minimum valuation. Such a component must either be lifted or be paired with another component at the same valuation floor, as classified later in Lemma 1H. $\square$

Remark 1 The role of Lemma 1F is valuation-theoretic. Each internal term

$$T_k = 3^{n-k} \left(2^{S_{k-1}} - 2^{2k-2} \right)$$

carries a visible positional factor 3^{n-k} . Thus different internal indices initially occupy different 3-adic valuation floors, unless the parenthetical factor supplies additional 3-adic lift. By the ultrametric property, if one active component has strictly smaller 3-adic valuation than every other active component, then the valuation of the whole sum is exactly that smaller valuation. Therefore a below-target component cannot be cancelled by components at higher 3-adic valuation; it must either be lifted to the target level or meet another active component at the same valuation floor.

Lemma 1G *Let*

$$T_k = 3^{n-k} \left(2^{S_{k-1}} - 2^{2k-2} \right)$$

be an active internal term, and define

$$d_k := |S_{k-1} - (2k - 2)|.$$

Assume that $d_k > 0$ is even. Let p be the local target valuation for T_k .

If

$$p \leq n - k,$$

then the visible positional factor 3^{n-k} already supplies the required valuation, and no additional LTE lift is required.

If

$$p > n - k,$$

then the term requires genuine additional 3-adic lift. In that case

$$d_k \geq 2 \cdot 3^{p-n+k-1}.$$

Proof Factoring out the smaller power of 2 from

$$2^{S_{k-1}} - 2^{2k-2},$$

the 3-adic contribution of the parenthetical difference is

$$v_3(2^{d_k} - 1).$$

Thus

$$v_3(T_k) = (n - k) + v_3(2^{d_k} - 1).$$

If

$$p \leq n - k,$$

then the visible positional valuation already gives

$$v_3(T_k) \geq n - k \geq p.$$

Hence no additional lower bound on d_k is needed.

Now assume

$$p > n - k.$$

Then T_k must gain additional valuation from the factor

$$2^{d_k} - 1.$$

The condition $v_3(T_k) \geq p$ becomes

$$(n - k) + v_3(2^{d_k} - 1) \geq p,$$

or equivalently

$$v_3(2^{d_k} - 1) \geq p - n + k.$$

Since d_k is even, the Lifting The Exponent lemma gives

$$v_3(2^{d_k} - 1) = v_3(2^2 - 1) + v_3\left(\frac{d_k}{2}\right).$$

Because

$$v_3(2^2 - 1) = v_3(3) = 1$$

and $v_3(2) = 0$, this becomes

$$v_3(2^{d_k} - 1) = 1 + v_3(d_k).$$

Therefore

$$1 + v_3(d_k) \geq p - n + k,$$

so

$$v_3(d_k) \geq p - n + k - 1.$$

Since $p > n - k$, the exponent

$$p - n + k - 1$$

is non-negative. Hence d_k must be divisible by

$$3^{p-n+k-1}.$$

Because d_k is positive and even, the smallest possible such value is

$$d_k = 2 \cdot 3^{p-n+k-1}.$$

Therefore

$$d_k \geq 2 \cdot 3^{p-n+k-1}.$$

□

Lemma 1H Permissible 3-adic trajectories. *Let the bifurcated left-hand side of Lemma 1F be*

$$\left(\sum_{k=2}^{n-1} T_k\right) + T_{\text{last}},$$

where

$$T_{\text{last}} = T_n + B.$$

If this left-hand side satisfies the required 3-adic valuation alignment with

$$(Z - 1)3^n,$$

then the valuation trajectory must occur through one of the following two mechanisms.

Mechanism 1: Direct lifting. *Every active term whose visible positional 3-power lies below the required target valuation*

$$n + v_3(Z - 1)$$

supplies the missing 3-adic valuation internally. Thus every below-target active term individually reaches the required valuation, and no lower valuation floor remains.

Mechanism 2: Hybrid cancellation-lift. *Some active terms first rise by direct internal lifting to a common valuation floor*

$$1 \leq p \leq n - 2.$$

At this common floor, cancellation among equal-valuation terms raises the valuation of their combined contribution. The same condition then repeats: each new below-target floor must either be lifted directly or removed by equal-floor cancellation.

Equivalently, the permissible exponent trajectory of Mechanism 2 has the following form. For some transition index q , the parity-inverting entry is

$$a_1 = 1 \quad \text{or} \quad a_1 \in \{3, 5, 7, \dots\}.$$

The coasting part satisfies

$$a_2, \dots, a_{q-1} \quad \text{are either equilibrium steps 2} \quad \text{or odd positive exponent steps.}$$

The bridge exponent satisfies

$$a_q = 1 \quad \text{or} \quad a_q \in \{3, 5, 7, \dots\}.$$

After the bridge, the lifted tail

$$a_{q+1}, \dots, a_{n-2}$$

must supply the positive 3-adic lift required by Lemma 1G. The remaining terminal exponents

$$a_{n-1}, \quad a_n$$

satisfy only the general lower bound

$$a_{n-1} \geq 1, \quad a_n \geq 1.$$

The endpoint

$$p = 1$$

is the no-lift endpoint of Mechanism 2. In this endpoint the lifted tail is shortened to one term, and the internal trajectory controlled by the mechanism is limited to

$$a_1, \dots, a_{n-2}.$$

The remaining terminal exponents a_{n-1} and a_n are therefore not restricted by this internal no-lift mechanism.

The range

$$1 < p < n - 2$$

is the genuine interior lifted cascade, while

$$p = n - 2$$

is the boundary endpoint of the same hybrid cancellation-lift mechanism.

Proof By Lemma 1F, the modified cycle equation gives

$$\left(\sum_{k=2}^{n-1} T_k \right) + T_{\text{last}} = (Z - 1)3^n.$$

Therefore the left-hand side must have the same 3-adic valuation as the right-hand side, namely

$$n + v_3(Z - 1).$$

Consider the active terms on the left-hand side. If every term whose visible positional 3-power lies below $n + v_3(Z - 1)$ gains the missing valuation internally, then every such term individually reaches the required valuation. This is precisely Mechanism 1.

It remains to consider the case in which some active term does not individually reach the required valuation. Let p be a below-target valuation floor reached by such active terms. A term at this floor cannot be unique. Indeed, if a unique term had the smallest valuation

floor, then the ultrametric valuation rule would force the whole left-hand side to have that same valuation, contradicting the required valuation

$$n + v_3(Z - 1).$$

Therefore every below-target valuation floor that is not removed by direct lifting must occur among at least two equal-valuation terms. Since 3-adic cancellation can occur only among terms of equal valuation, the terms at the common floor p must cancel after division by 3^p . This raises the valuation of their combined contribution above p .

The parity restrictions on this cancellation-lift path are those of Lemma 1E. A parity-inverting step is supplied by an odd exponent. The minimal such step is the negative perturbation $a_i = 1$, while the non-minimal possibilities are odd positive exponents 3, 5, 7, ... Equilibrium steps $a_i = 2$ preserve the existing parity. Once the bridge has occurred, the lifted tail must supply the positive deviation required by Lemma 1G. This gives exactly the trajectory displayed in the statement:

$$\begin{aligned} a_1 &= 1 \text{ or odd positive,} & a_2, \dots, a_{q-1} &= 2 \text{ or odd positive,} \\ a_q &= 1 \text{ or odd positive,} & a_{q+1}, \dots, a_{n-2} &\text{ satisfy the LTE lift.} \end{aligned}$$

The terminal exponents a_{n-1} and a_n lie outside the internally controlled chain in the no-lift endpoint, and otherwise retain only the general bound $a_i \geq 1$.

The same equal-floor cancellation argument applies at each subsequent valuation level. If a new below-target floor is unique, then the required valuation is impossible. If it is not unique, it must again be removed by equal-floor cancellation. This iterative equal-floor cancellation is the hybrid cancellation-lift trajectory of Mechanism 2.

Since the internal trajectory governed by this mechanism runs through the truncated internal term chain, the possible floor satisfies

$$1 \leq p \leq n - 2.$$

The endpoint $p = 1$ gives the no-lift endpoint; the range $1 < p < n - 2$ gives the genuine lifted cascade; and $p = n - 2$ gives the boundary endpoint. These are not separate mechanisms, but the endpoint and interior cases of the same hybrid cancellation-lift mechanism. $\square$

Remark 2 Lemma 1H has a strictly local scope: it classifies the possible 3-adic routes by which the bifurcated left-hand side of Lemma 1F can reach the cycle-forced target valuation. The largest visible positional 3-power occurring among the internal left-hand-side terms is below the target valuation; in particular, the internal positional range is bounded by the truncated chain ending before the terminal boundary block. Therefore any active below-target component must gain additional 3-adic valuation before the full left-hand side can align with

$$v_3((1 - Z)3^n) = n + v_3(Z - 1).$$

There are only two ways this gain can occur. First, an individual component may acquire the missing 3-adic valuation inside its multiplicative factor

$$2^{S_{k-1}} - 2^{2k-2};$$

this is Mechanism 1, direct lifting. Second, two or more active components may first reach the same 3-adic valuation floor and then cancel after division by that common power of 3; this is Mechanism 2, hybrid cancellation-lift.

No third mechanism is available. By the ultrametric property, terms at unequal 3-adic valuation floors cannot cancel the unique lowest floor: if one active component has strictly smaller valuation than all the others, then the valuation of the whole sum is exactly that smaller valuation. Thus every below-target floor must either be removed by direct internal lift or by equal-floor cancellation. This is the exhaustive dichotomy used in the subsequent branch analysis.

Lemma 1I *Pure negative no-lift endpoint.* In the $p = 1$ no-lift endpoint of Mechanism 2, every admissible pure negative terminal trajectory gives a non-integral cycle ratio

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}}.$$

Consequently, the pure negative no-lift endpoint cannot realize a non-trivial positive integer cycle.

Proof In the $p = 1$ no-lift endpoint of Mechanism 2, the internal trajectory controlled by the mechanism is limited to

$$a_1, \dots, a_{n-2}.$$

In the pure negative no-lift endpoint, the internally controlled trajectory is

$$a_1 = 1, \quad a_2 = a_3 = \dots = a_{n-2} = 2.$$

The remaining terminal exponents a_{n-1} and a_n are not restricted by this internal no-lift mechanism. Thus the pure negative no-lift branch can contain negative perturbations only at

$$a_1, \quad a_{n-1}, \quad a_n.$$

The exponent a_{n-1} is visible to the numerator, while a_n affects only the denominator.

Case I: Only a_1 is negatively perturbed. Assume

$$a_1 = 1, \quad a_2 = \dots = a_n = 2.$$

Then

$$S_n = 2n - 1,$$

and hence

$$D_{\text{new}} = 2^{2n-1} - 3^n.$$

The numerator is

$$N_{\text{new}} = \sum_{k=0}^{n-1} 3^{n-1-k} 2^{S_k}.$$

For $k = 0$, the prefix is unchanged. For every $k \geq 1$, the first negative perturbation has occurred, so

$$S_k = 2k - 1.$$

Therefore

$$\Delta N_{\text{actual}} = \sum_{k=1}^{n-1} 3^{n-1-k} (2^{2k-1} - 2^{2k}) = - \sum_{k=1}^{n-1} 3^{n-1-k} 2^{2k-1}.$$

Evaluating this finite geometric progression gives

$$\Delta N_{\text{actual}} = - (2^{2n-1} - 2 \cdot 3^{n-1}).$$

Since

$$N_{\text{eq}} = 2^{2n} - 3^n,$$

we obtain

$$N_{\text{new}} = 2^{2n-1} - 3^n + 2 \cdot 3^{n-1}.$$

Thus

$$Z = \frac{2^{2n-1} - 3^n + 2 \cdot 3^{n-1}}{2^{2n-1} - 3^n} = 1 + \frac{2 \cdot 3^{n-1}}{2^{2n-1} - 3^n}.$$

If Z were an integer, then

$$2^{2n-1} - 3^n \mid 2 \cdot 3^{n-1}.$$

But $2^{2n-1} - 3^n$ is odd and is not divisible by 3. Hence it is coprime to $2 \cdot 3^{n-1}$. Therefore it would have to divide 1. In the valid positive-denominator non-trivial range,

$$2^{2n-1} - 3^n > 1.$$

This is impossible. Hence

$$Z \notin \mathbb{Z}.$$

Case II: Negative perturbations at a_1 and a_{n-1} . Assume

$$a_1 = 1, \quad a_{n-1} = 1,$$

and all remaining exponents equal 2, including $a_n = 2$.

For the numerator-deviation term, write

$$U_j = 3^{n-1-j} \left(2^{S_j} - 2^{2j} \right), \quad 0 \leq j \leq n-1.$$

Then

$$\Delta N_{\text{actual}} = \sum_{j=0}^{n-1} U_j.$$

For $1 \leq j \leq n-2$, only the first negative perturbation has occurred, so

$$S_j = 2j - 1.$$

At the final numerator prefix,

$$S_{n-1} = 2n - 4.$$

Since $a_n = 2$, the denominator exponent is

$$S_n = 2n - 2.$$

The initial term satisfies $U_0 = 0$. For $1 \leq j \leq n-2$,

$$U_j = 3^{n-1-j} \left(2^{2j-1} - 2^{2j} \right) = -3^{n-1-j} 2^{2j-1}.$$

Thus

$$-\sum_{j=1}^{n-2} 3^{n-1-j} 2^{2j-1} = -6 \left(4^{n-2} - 3^{n-2} \right).$$

The final numerator-deviation term is

$$U_{n-1} = 2^{2n-4} - 2^{2n-2} = -3 \cdot 4^{n-2}.$$

Therefore

$$\Delta N_{\text{actual}} = 6 \cdot 3^{n-2} - 9 \cdot 4^{n-2}.$$

Since

$$N_{\text{eq}} = 4^n - 3^n,$$

we obtain

$$N_{\text{new}} = 7 \cdot 4^{n-2} - 3^{n-1}.$$

The denominator is

$$D_{\text{new}} = 4^{n-1} - 3^n.$$

Hence

$$Z = \frac{7 \cdot 4^{n-2} - 3^{n-1}}{4^{n-1} - 3^n}.$$

Let

$$A = 4^{n-2}, \quad C = 3^{n-2}.$$

Then

$$N_{\text{new}} = 7A - 3C, \quad D_{\text{new}} = 4A - 9C.$$

Assume, for contradiction, that $D_{\text{new}} \mid N_{\text{new}}$. Then

$$D_{\text{new}} \mid N_{\text{new}} - D_{\text{new}} = 3A + 6C = 3(A + 2C).$$

Since

$$D_{\text{new}} \equiv A \equiv 1 \pmod{3},$$

we have

$$\gcd(D_{\text{new}}, 3) = 1.$$

Hence

$$D_{\text{new}} \mid A + 2C.$$

Therefore

$$D_{\text{new}} \mid 4(A + 2C) - D_{\text{new}} = 17C.$$

Since $C = 3^{n-2}$ and

$$\gcd(D_{\text{new}}, C) = 1,$$

it follows that

$$D_{\text{new}} \mid 17.$$

In the valid positive-denominator range, this is impossible. Therefore

$$Z \notin \mathbb{Z}.$$

Case III: Final denominator-edge perturbation $a_n = 1$. The final exponent a_n does not occur in the numerator, since

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$$

depends only on the prefix sums up to S_{n-1} . Thus a_n modifies only the denominator.

Subcase III-A: $a_1 = 1$, $a_{n-1} = 2$, and $a_n = 1$. From Case I, the numerator is

$$N_{\text{new}} = 2^{2n-1} - 3^n + 2 \cdot 3^{n-1}.$$

The total exponent is

$$S_n = 2n - 2,$$

so

$$D_{\text{new}} = 2^{2n-2} - 3^n.$$

Therefore

$$Z = \frac{2^{2n-1} - 3^n + 2 \cdot 3^{n-1}}{2^{2n-2} - 3^n}.$$

Assume, for contradiction, that $Z \in \mathbb{Z}$. Then

$$D_{\text{new}} \mid N_{\text{new}} - 2D_{\text{new}}.$$

But

$$N_{\text{new}} - 2D_{\text{new}} = 5 \cdot 3^{n-1}.$$

Since

$$D_{\text{new}} \equiv 1 \pmod{3},$$

we have

$$\gcd(D_{\text{new}}, 3) = 1.$$

Hence

$$D_{\text{new}} \mid 5.$$

In the valid positive-denominator non-trivial range, this is impossible. Therefore

$$Z \notin \mathbb{Z}.$$

Subcase III-B: $a_1 = 1$, $a_{n-1} = 1$, and $a_n = 1$. From Case II, the numerator is

$$N_{\text{new}} = 7 \cdot 4^{n-2} - 3^{n-1}.$$

Now

$$S_n = 2n - 3,$$

so

$$D_{\text{new}} = 2^{2n-3} - 3^n.$$

Writing

$$A = 4^{n-2}, \quad C = 3^{n-2},$$

we have

$$N_{\text{new}} = 7A - 3C, \quad D_{\text{new}} = 2A - 9C.$$

Assume, for contradiction, that $D_{\text{new}} \mid N_{\text{new}}$. Then

$$D_{\text{new}} \mid N_{\text{new}} - 3D_{\text{new}} = A + 24C.$$

Therefore

$$D_{\text{new}} \mid 2(A + 24C) - D_{\text{new}} = 57C.$$

Since

$$D_{\text{new}} \equiv 2A \equiv 2 \pmod{3},$$

we have

$$\gcd(D_{\text{new}}, 3) = 1.$$

Also $C = 3^{n-2}$, so

$$\gcd(D_{\text{new}}, C) = 1.$$

Hence

$$D_{\text{new}} \mid 57.$$

Since $\gcd(D_{\text{new}}, 3) = 1$, this implies

$$D_{\text{new}} \mid 19.$$

In the valid positive-denominator non-trivial range, this is impossible. Therefore

$$Z \notin \mathbb{Z}.$$

All pure negative terminal configurations in the $p = 1$ no-lift endpoint therefore give non-integral cycle ratios. Hence this endpoint cannot produce a non-trivial positive integer cycle. $\square$

Corollary 1I-1 Closure of the pure negative no-lift branch. *In the no-lift pure negative regime, the only possible full exponent trajectory can contain negative perturbations at*

$$a_1, \quad a_{n-1}, \quad a_n.$$

Among these, only a_1 and a_{n-1} are visible to the numerator, because the numerator expansion terminates at S_{n-1} . The final negative perturbation $a_n = 1$, if present, affects only the denominator.

Consequently, the pure negative no-lift branch has exactly the terminal subcases enumerated in Lemma 1I. In every such subcase, the cycle ratio

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}}$$

is non-integral. Hence the pure negative no-lift branch cannot produce a non-trivial integer cycle.

Proof In the no-lift pure negative regime, Lemma 1I fixes the internal trajectory as

$$a_1 = 1, \quad a_2 = a_3 = \cdots = a_{n-2} = 2.$$

Thus no additional negative perturbation can occur in the internal range.

The numerator is

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j},$$

so the largest prefix exponent appearing in the numerator is S_{n-1} . Therefore a_n does not enter the numerator. The only remaining numerator-visible terminal negative perturbation is $a_{n-1} = 1$, while $a_n = 1$, if present, changes only the denominator through S_n .

Hence the no-lift pure negative branch is exhausted by the terminal cases covered in Lemma 1I:

$$(a_{n-1}, a_n) = (2, 2), \quad (1, 2), \quad (2, 1), \quad (1, 1).$$

Lemma 1I proves in each case that

$$\frac{N_{\text{new}}}{D_{\text{new}}} \notin \mathbb{Z}.$$

Therefore the pure negative no-lift branch violates the integer cycle condition and cannot close a non-trivial Collatz cycle. $\square$

Lemma 1J *Mechanism-2 ratio ceiling for the maximal-ratio minimal-lift representative.* Assume that the modified cycle equation enters the interior lifted Mechanism-2 hybrid cancellation-lift branch of Lemma 1H. Let

$$1 < p < n - 2$$

be the internal valuation floor, and write

$$q = n - p - 1.$$

Let the lifted tail be minimal in the sense that equality is attained in the LTE lower bound of Lemma 1G. Assume also that the final exponent appearing in the numerator is minimized, namely

$$a_{n-1} = 1.$$

Then the cycle ratio satisfies

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}} < 5.$$

Consequently, this maximal-ratio minimal-lift representative cannot realize a non-trivial positive integer cycle ratio.

Proof Let

$$S_k = \sum_{i=1}^k a_i, \quad e_k = S_k - 2k$$

be the cumulative exponent sum and its signed deviation from the equilibrium prefix. For the numerator-deviation terms, write

$$T_k = 3^{n-k} \left(2^{S_{k-1}} - 2^{2k-2} \right), \quad 1 \leq k \leq n.$$

Since

$$S_{k-1} = 2k - 2 + e_{k-1},$$

we have

$$T_k = 3^{n-k} 2^{2k-2} \left(2^{e_{k-1}} - 1 \right).$$

Therefore

$$\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}} = \sum_{k=1}^n T_k.$$

Among the Mechanism-2-permitted parity-inverting choices, the maximal-ratio representative is obtained by taking the minimal allowed exponents. Thus the parity-inverting and terminal numerator-visible choices are taken to be negative perturbations where permitted. Any replacement of such a permitted 1 by an odd positive exponent, or any increase in the lifted tail, is a componentwise positive exponent inflation and therefore decreases the cycle ratio by Lemma 1K. Hence it is enough to estimate the minimal lifted representative, where the lifted tail attains equality in the LTE lower bound of Lemma 1G.

By Mechanism 2 of Lemma 1H, the no-lift coasting segment is followed by a bridge perturbation and then by a lifted tail. Let

$$q = n - p - 1.$$

Since $p \geq 2$ and the bridge index satisfies $q \geq 2$, this branch has

$$n \geq 5.$$

The maximal-ratio minimal-lift exponent profile is as follows. First,

$$a_1 = 1,$$

so

$$e_1 = -1.$$

During the coasting phase,

$$a_2 = \cdots = a_{q-1} = 2,$$

and hence

$$e_1 = e_2 = \cdots = e_{q-1} = -1.$$

At the bridge index,

$$a_q = 1,$$

so

$$e_q = e_{q-1} + a_q - 2 = -2.$$

After the bridge, the minimal lifted tail is obtained by taking equality in the quantitative lower bound of Lemma 1G. Thus, for

$$q + 1 \leq j \leq n - 2,$$

the signed deviations are

$$e_j = 2 \cdot 3^{j-q-1}.$$

Equivalently, since $q = n - p - 1$,

$$e_j = 2 \cdot 3^{p+j-n}.$$

In particular,

$$e_{n-2} = 2 \cdot 3^{p-2}.$$

The corresponding lifted-tail exponents are obtained from

$$a_j = 2 + e_j - e_{j-1}.$$

Thus

$$a_{q+1} = 2 + 2 - (-2) = 6,$$

and for $q + 2 \leq j \leq n - 2$,

$$a_j = 2 + 2 \cdot 3^{j-q-1} - 2 \cdot 3^{j-q-2} = 2 + 4 \cdot 3^{j-q-2}.$$

The exponent a_n does not occur in the numerator, because

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$$

uses only prefix sums up to S_{n-1} . However, a_n does occur in the denominator

$$D_{\text{new}} = 2^{S_n} - 3^n.$$

Therefore, for this fixed numerator profile, the ratio

$$\frac{N_{\text{new}}}{D_{\text{new}}}$$

is largest when the denominator is smallest, namely when

$$a_n = 1.$$

Thus it is enough to prove $Z < 5$ in the extremal terminal case

$$a_{n-1} = 1, \quad a_n = 1.$$

Step 1: Derivation of GP_1 . Split the numerator deviation as

$$\Delta N_{\text{actual}} = GP_1 + T_{q+1} + GP_2,$$

where

$$GP_1 = \sum_{k=1}^q T_k, \quad GP_2 = \sum_{k=q+2}^n T_k.$$

Since $T_1 = 0$, and since $e_{k-1} = -1$ for $2 \leq k \leq q$, we obtain

$$T_k = 3^{n-k} 2^{2k-2} (2^{-1} - 1) = -3^{n-k} 2^{2k-3}$$

for $2 \leq k \leq q$. Hence

$$GP_1 = - \sum_{k=2}^q 3^{n-k} 2^{2k-3}.$$

This is a finite geometric progression. Its first positive magnitude term is

$$2 \cdot 3^{n-2},$$

and its common ratio is

$$\frac{4}{3}.$$

Therefore

$$\sum_{k=2}^q 3^{n-k} 2^{2k-3} = 2 \cdot 3^{n-2} \left[\frac{(4/3)^{q-1} - 1}{(4/3) - 1} \right].$$

Since

$$(4/3) - 1 = \frac{1}{3},$$

this becomes

$$\sum_{k=2}^q 3^{n-k} 2^{2k-3} = 6 \left(4^{q-1} 3^{n-q-1} - 3^{n-2} \right).$$

Thus

$$GP_1 = -6 \left(4^{q-1} 3^{n-q-1} - 3^{n-2} \right).$$

Using $q = n - p - 1$, this may also be written as

$$GP_1 = -6 \left(4^{n-p-2} 3^p - 3^{n-2} \right).$$

In particular,

$$GP_1 < 0.$$

Step 2: Derivation of T_{q+1} . At the bridge output,

$$e_q = -2.$$

Therefore

$$T_{q+1} = 3^{n-q-1} 2^{2q} \left(2^{-2} - 1 \right).$$

Since

$$2^{-2} - 1 = \frac{1}{4} - 1 = -\frac{3}{4},$$

we get

$$T_{q+1} = -3^{n-q} 2^{2q-2}.$$

Using $q = n - p - 1$, this is

$$T_{q+1} = -3^{p+1} 4^{n-p-2}.$$

Hence

$$T_{q+1} < 0.$$

Step 3: Derivation and bound of GP_2 . For $q + 2 \leq k \leq n - 1$, the relevant deviation is

$$e_{k-1} = 2 \cdot 3^{k-q-2}.$$

Thus

$$T_k = 3^{n-k} 2^{2k-2} \left(2^{2 \cdot 3^{k-q-2}} - 1 \right) \quad (q + 2 \leq k \leq n - 1).$$

For the final numerator-visible term T_n , we use

$$e_{n-1} = e_{n-2} + a_{n-1} - 2 = 2 \cdot 3^{p-2} - 1.$$

Therefore

$$T_n = 2^{2n-2} \left(2^{2 \cdot 3^{p-2}-1} - 1 \right).$$

Consequently,

$$GP_2 = \sum_{k=q+2}^{n-1} 3^{n-k} 2^{2k-2} \left(2^{2 \cdot 3^{k-q-2}} - 1 \right) + 2^{2n-2} \left(2^{2 \cdot 3^{p-2}-1} - 1 \right).$$

Now set

$$L = 2^{S_n}.$$

Since all terms in GP_2 have positive deviation, each satisfies

$$T_k < 3^{n-k} 2^{S_{k-1}}.$$

Thus GP_2 is bounded above by the corresponding numerator-visible terms.

For $k = n$,

$$3^{n-n} 2^{S_{n-1}} = 2^{S_{n-1}}.$$

Since $a_n = 1$,

$$S_{n-1} = S_n - 1,$$

and therefore

$$2^{S_{n-1}} = \frac{L}{2}.$$

For $k = n - 1$,

$$3^{n-(n-1)} 2^{S_{n-2}} = 3 \cdot 2^{S_{n-2}}.$$

Since $a_{n-1} = a_n = 1$,

$$S_{n-2} = S_n - 2,$$

and hence

$$3 \cdot 2^{S_{n-2}} = \frac{3L}{4}.$$

For $k = n - 2$,

$$3^{n-(n-2)} 2^{S_{n-3}} = 9 \cdot 2^{S_{n-3}}.$$

The exponent gap from S_{n-3} to S_n is

$$a_{n-2} + a_{n-1} + a_n.$$

In the minimal lifted tail,

$$a_{n-2} \geq 6,$$

while

$$a_{n-1} = a_n = 1.$$

Therefore

$$S_{n-3} \leq S_n - 8,$$

and so

$$9 \cdot 2^{S_{n-3}} \leq \frac{9L}{256}.$$

Moving backward through the earlier lifted-tail terms, the ratio of consecutive numerator-visible terms is

$$\frac{3^{n-(k-1)} 2^{S_{k-2}}}{3^{n-k} 2^{S_{k-1}}} = \frac{3}{2^{a_{k-1}}}.$$

Inside the lifted tail, every such exponent satisfies

$$a_{k-1} \geq 6.$$

Therefore the backward ratio is at most

$$\frac{3}{64}.$$

Hence the entire earlier lifted part is bounded by

$$\frac{9L}{256} \left(1 + \frac{3}{64} + \left(\frac{3}{64} \right)^2 + \cdots \right).$$

Summing this geometric series gives

$$\frac{9L}{256} \cdot \frac{1}{1 - 3/64} = \frac{9L}{244}.$$

Combining the final term, the penultimate term, and the earlier lifted-tail bound yields

$$GP_2 < \frac{L}{2} + \frac{3L}{4} + \frac{9L}{244}.$$

Since

$$\frac{1}{2} + \frac{3}{4} + \frac{9}{244} = \frac{157}{122},$$

we obtain

$$GP_2 < \frac{157}{122}L.$$

Step 4: Derivation of ΔN_{actual} . From the decomposition,

$$\Delta N_{\text{actual}} = GP_1 + T_{q+1} + GP_2.$$

Steps 1 and 2 showed that

$$GP_1 < 0, \quad T_{q+1} < 0.$$

Therefore

$$\Delta N_{\text{actual}} < GP_2.$$

Using the bound from Step 3,

$$\Delta N_{\text{actual}} < \frac{157}{122}L.$$

Step 5: Derivation of S_n . We now compute the total exponent sum in the extremal case

$$a_{n-1} = 1, \quad a_n = 1.$$

The exponent profile gives

$$S_n = 1 + 2(q-2) + 1 + \left[6 + \sum_{r=2}^{p-1} (2 + 4 \cdot 3^{r-2}) \right] + 1 + 1,$$

where the finite sum is empty when $p = 2$. Here:

- the first 1 is $a_1 = 1$;
- the block $2(q-2)$ is $a_2 = \dots = a_{q-1} = 2$;
- the next 1 is the bridge exponent $a_q = 1$;
- the bracketed expression is the lifted-tail sum from a_{q+1} to a_{n-2} ;
- the final $1 + 1$ is $a_{n-1} + a_n$.

Evaluate the lifted-tail sum:

$$6 + \sum_{r=2}^{p-1} (2 + 4 \cdot 3^{r-2}) = 6 + 2(p-2) + 4 \sum_{r=2}^{p-1} 3^{r-2}.$$

Letting $s = r - 2$, we get

$$\sum_{r=2}^{p-1} 3^{r-2} = \sum_{s=0}^{p-3} 3^s = \frac{3^{p-2} - 1}{3 - 1}.$$

Thus

$$4 \sum_{r=2}^{p-1} 3^{r-2} = 4 \cdot \frac{3^{p-2} - 1}{2} = 2(3^{p-2} - 1).$$

Therefore

$$6 + \sum_{r=2}^{p-1} (2 + 4 \cdot 3^{r-2}) = 6 + 2(p-2) + 2(3^{p-2} - 1),$$

so

$$6 + \sum_{r=2}^{p-1} (2 + 4 \cdot 3^{r-2}) = 2p + 2 \cdot 3^{p-2}.$$

Since

$$q = n - p - 1,$$

we obtain

$$S_n = 1 + 2(q - 2) + 1 + 2p + 2 \cdot 3^{p-2} + 1 + 1.$$

This simplifies to

$$S_n = 2q + 2p + 2 \cdot 3^{p-2}.$$

Substituting $q = n - p - 1$, we get

$$S_n = 2(n - p - 1) + 2p + 2 \cdot 3^{p-2},$$

and hence

$$S_n = 2n - 2 + 2 \cdot 3^{p-2}.$$

Step 6: Derivation of D_{new} . By definition,

$$D_{\text{new}} = 2^{S_n} - 3^n.$$

Since

$$L = 2^{S_n},$$

we have

$$D_{\text{new}} = L - 3^n.$$

From Step 5,

$$S_n = 2n - 2 + 2 \cdot 3^{p-2}.$$

Because $p \geq 2$,

$$2 \cdot 3^{p-2} \geq 2,$$

and therefore

$$S_n \geq 2n.$$

Thus

$$\frac{3^n}{L} = \frac{3^n}{2^{S_n}} \leq \frac{3^n}{2^{2n}} = \left(\frac{3}{4}\right)^n.$$

Since $n \geq 5$,

$$\left(\frac{3}{4}\right)^n \leq \left(\frac{3}{4}\right)^5 = \frac{243}{1024} < \frac{1}{4}.$$

Hence

$$3^n < \frac{L}{4}.$$

Therefore

$$D_{\text{new}} = L - 3^n > \frac{3L}{4}.$$

Step 7: Derivation of Z . The perturbed numerator is

$$N_{\text{new}} = N_{\text{eq}} + \Delta N_{\text{actual}}.$$

Since

$$N_{\text{eq}} = 2^{2n} - 3^n < 2^{2n},$$

and since $S_n \geq 2n$, we have

$$N_{\text{eq}} < L.$$

Using Step 4,

$$\Delta N_{\text{actual}} < \frac{157}{122}L.$$

Therefore

$$N_{\text{new}} < L + \frac{157}{122}L = \frac{279}{122}L.$$

Combining this numerator bound with the denominator bound from Step 6 gives

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}} < \frac{\frac{279}{122}L}{\frac{3}{4}L}.$$

Thus

$$Z < \frac{279}{122} \cdot \frac{4}{3} = \frac{186}{61}.$$

Since

$$\frac{186}{61} < 5,$$

we conclude that

$$Z < 5.$$

This proves the ratio ceiling in the denominator-minimized terminal case $a_n = 1$. If $a_n > 1$, then the numerator is unchanged while

$$D_{\text{new}} = 2^{S_n} - 3^n$$

increases. Hence the ratio is even smaller. Therefore every minimal $C = 1$ Mechanism-2 lifted-tail profile with $a_{n-1} = 1$ satisfies

$$Z < 5.$$

By Constraint 3, a non-trivial positive integer cycle ratio must satisfy

$$Z \geq 5.$$

Hence this maximal-ratio minimal-lift representative cannot realize a non-trivial positive integer cycle. $\square$

Lemma 1K *Positive Exponent Inflation Decreases the Cycle Ratio. Let*

$$a = (a_1, \dots, a_n)$$

be any exponent profile with

$$D(a) = 2^S - 3^n > 0, \quad S = \sum_{i=1}^n a_i.$$

Let

$$\hat{a} = (\hat{a}_1, \dots, \hat{a}_n)$$

be another exponent profile satisfying

$$\hat{a}_i \geq a_i \quad (1 \leq i \leq n),$$

with at least one strict inequality. Define

$$\hat{S} = \sum_{i=1}^n \hat{a}_i, \quad R = \hat{S} - S.$$

Then

$$\frac{N(\hat{a})}{D(\hat{a})} < \frac{N(a)}{D(a)}.$$

Thus any componentwise positive exponent inflation makes the denominator grow faster than the numerator.

Proof For the original profile a , write

$$S_j = \sum_{i=1}^j a_i, \quad S = S_n.$$

Then

$$N(a) = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}, \quad D(a) = 2^S - 3^n.$$

For the inflated profile $\hat{a}$, write

$$\hat{S}_j = \sum_{i=1}^j \hat{a}_i, \quad \hat{S} = \hat{S}_n.$$

Then

$$N(\hat{a}) = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{\hat{S}_j}, \quad D(\hat{a}) = 2^{\hat{S}} - 3^n.$$

Since

$$\hat{a}_i \geq a_i$$

for every i , the prefix increase satisfies

$$0 \leq \hat{S}_j - S_j \leq \hat{S} - S = R \quad (0 \leq j \leq n-1).$$

Hence

$$2^{\hat{S}_j} = 2^{S_j} 2^{\hat{S}_j - S_j} \leq 2^{S_j} 2^R.$$

Therefore

$$N(\hat{a}) \leq 2^R \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j} = 2^R N(a).$$

Now compare the denominators:

$$D(\hat{a}) = 2^{\hat{S}} - 3^n = 2^{S+R} - 3^n.$$

Since

$$2^S = D(a) + 3^n,$$

we obtain

$$D(\hat{a}) = 2^R (D(a) + 3^n) - 3^n.$$

Therefore

$$D(\hat{a}) = 2^R D(a) + (2^R - 1)3^n.$$

Because at least one exponent is increased, we have

$$R > 0.$$

Thus

$$(2^R - 1)3^n > 0,$$

and consequently

$$D(\hat{a}) > 2^R D(a).$$

Combining the numerator and denominator estimates gives

$$N(\hat{a}) \leq 2^R N(a), \quad D(\hat{a}) > 2^R D(a).$$

Since $D(a) > 0$, it follows that

$$\frac{N(\hat{a})}{D(\hat{a})} < \frac{2^R N(a)}{2^R D(a)} = \frac{N(a)}{D(a)}.$$

Hence componentwise positive exponent inflation strictly decreases the cycle ratio. In particular, once a maximal-ratio representative has been checked, all profiles obtained from it by replacing permitted 1's with odd positive exponents, by increasing lifted-tail exponents, or by increasing terminal exponents are automatically smaller in ratio. $\square$

Theorem 1 *Non-Existence of Non-Trivial Integer Cycles.* *For the accelerated Collatz operator, no non-trivial positive integer cycle exists.*

Proof Assume, for contradiction, that a non-trivial positive integer cycle exists. Let

$$a = (a_1, \dots, a_n)$$

be its vector of division exponents, and let

$$S_0 = 0, \quad S_j = \sum_{i=1}^j a_i, \quad S = S_n.$$

The fundamental cycle equation gives

$$Z = \frac{N_{\text{new}}}{D_{\text{new}}},$$

where

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}, \quad D_{\text{new}} = 2^S - 3^n.$$

Since the cycle is positive,

$$D_{\text{new}} > 0.$$

Since the cycle is assumed to be an integer cycle,

$$Z \in \mathbb{Z}^+.$$

By Lemma 1A, the equilibrium profile

$$a_1 = \dots = a_n = 2$$

gives only the trivial cycle. Hence a non-trivial cycle must arise from a non-equilibrium exponent profile.

We first close the neutral branch. If

$$N_{\text{new}} = D_{\text{new}},$$

then, since $N_{\text{eq}} = D_{\text{eq}}$, we have

$$\Delta N_{\text{actual}} = \Delta D.$$

By Lemma 3D, this equality is possible only for the equilibrium profile

$$a_1 = \dots = a_n = 2.$$

This profile is already handled by Lemma 1A. Therefore a non-equilibrium non-trivial cycle cannot occur in the neutral branch.

Thus every remaining candidate satisfies either

$$N_{\text{new}} < D_{\text{new}}$$

or

$$N_{\text{new}} > D_{\text{new}}.$$

The first case gives $0 < Z < 1$, impossible for $Z \in \mathbb{Z}^+$. Therefore the only remaining case is

$$N_{\text{new}} > D_{\text{new}}.$$

By Constraint 3, this numerator-dominant branch requires

$$Z \geq 5.$$

Every non-equilibrium exponent profile is pure positive, pure negative, or mixed.

1. Pure positive perturbations. If every perturbation from equilibrium is non-negative and at least one is positive, Lemma 1D gives

$$D_{\text{new}} > N_{\text{new}}.$$

This is the already excluded branch $N_{\text{new}} < D_{\text{new}}$. Hence no non-trivial integer cycle occurs in the pure positive branch.

2. Pure negative perturbations. In the pure negative branch, every negative perturbation has the form

$$a_i = 1,$$

since all division exponents satisfy $a_i \geq 1$.

If the bifurcated left-hand side of Lemma 1F fails

$$v_3(T_{\text{last}}) \geq 1,$$

then the required 3-adic valuation alignment cannot hold, so the cycle equation cannot close.

Otherwise Lemma 1H applies. In the pure negative setting, the only surviving cancellation-lift endpoint is the no-lift endpoint $p = 1$ of Mechanism 2. By Lemma 1I and Corollary 1I-1, every admissible terminal configuration in this endpoint gives

$$Z \notin \mathbb{Z}.$$

This contradicts the integer cycle condition.

3. Mixed perturbations. It remains to consider mixed perturbations.

If

$$v_3(T_{\text{last}}) = 0,$$

then Lemma 1F gives an immediate 3-adic contradiction. Therefore any mixed cycle candidate must enter one of the permissible trajectories of Lemma 1H.

There are two mechanisms.

Mechanism 1: Direct lifting. In the direct-lifting trajectory, every below-target active term supplies the missing 3-adic valuation internally. This is not a pure positive perturbation branch in general; terminal or parity-producing exponents may still be equal to 1. Rather, the required internal repair is achieved by componentwise positive exponent lift relative to the corresponding lower-ratio representative. By Lemma 1K, such positive exponent inflation strictly decreases the cycle ratio. Therefore a direct-lifting profile cannot realize a non-trivial integer cycle satisfying

$$Z \geq 5.$$

Mechanism 2: Hybrid cancellation-lift. In the hybrid cancellation-lift trajectory, active terms first reach a common valuation floor and then rise by equal-floor cancellation. By Lemma 1H,

$$1 \leq p \leq n - 2.$$

The endpoint

$$p = 1$$

is the no-lift endpoint already handled by Lemma 1I and Corollary 1I-1.

It remains to consider the lifted Mechanism-2 range

$$2 \leq p \leq n - 2.$$

This range splits into the interior range

$$1 < p < n - 2$$

and the boundary value

$$p = n - 2.$$

For the interior range, Lemma 1J proves that the maximal-ratio minimal lifted-tail representative satisfies

$$Z < 5.$$

If $p = n - 2$, then

$$q = n - p - 1 = 1.$$

This is the boundary lifted case in which the lifted tail begins immediately after the initial parity-inverting term. It is obtained from the corresponding lower-ratio representative by componentwise positive exponent inflation, and therefore Lemma 1K applies. Hence this boundary case cannot increase the ratio into the required numerator-dominant integer-cycle range

$$Z \geq 5.$$

Any further positive exponent inflation, including replacing permitted 1's by odd positive exponents, increasing lifted-tail exponents, or increasing terminal exponents such as a_{n-1} and a_n , decreases the cycle ratio by Lemma 1K. Therefore every lifted Mechanism-2 profile is excluded. This contradicts the required numerator-dominant condition $Z \geq 5$.

The pure positive, pure negative, and mixed branches exhaust all non-equilibrium exponent profiles. Each branch is impossible: it either falls below the denominator, violates the required 3-adic valuation alignment, yields a non-integral cycle ratio, or satisfies $Z < 5$, contradicting the necessary numerator-dominant condition $Z \geq 5$.

Therefore no non-trivial positive integer cycle exists for the accelerated Collatz operator. $\square$

5 Conclusion

The preceding theorem eliminates every non-equilibrium exponent profile in the accelerated Collatz cycle equation. Pure positive, pure negative, and mixed perturbations exhaust all possible departures from the equilibrium state, and each branch is ruled out by denominator dominance, non-integrality, failed 3-adic valuation alignment, or the ratio bound $Z < 5$.

Therefore the only positive integer cycle that remains is the equilibrium cycle identified in Lemma 1A. In the original Collatz map, this is precisely the trivial cycle

$$1 \mapsto 4 \mapsto 2 \mapsto 1.$$

Hence, for positive integers, the Collatz map admits no non-trivial integer cycles; the cycle 1-4-2-1 is the unique positive integer cycle.

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